

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS ENGINEERING,
SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES



Evolution of Natural Language Processing From Rule-Based Systems to Modern NLP



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What is Natural Language Processing (NLP)?

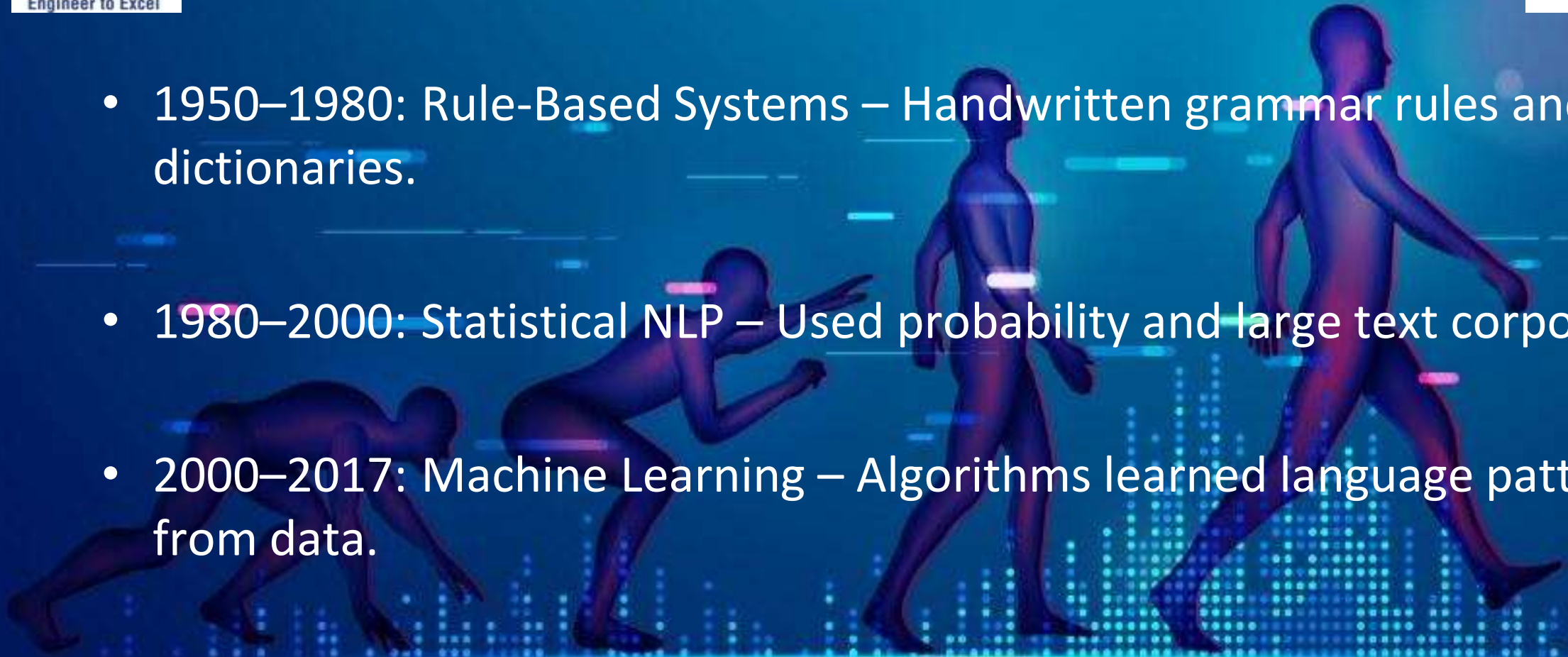
Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, generate and respond to human language.

- Combines AI, Machine Learning and Linguistics.
- Converts unstructured text into meaningful information.
- Used in chatbots, translation, voice assistants and search engines.



Evolution of NLP

- 1950–1980: Rule-Based Systems – Handwritten grammar rules and dictionaries.
- 1980–2000: Statistical NLP – Used probability and large text corpora.
- 2000–2017: Machine Learning – Algorithms learned language patterns from data.
- 2017–Present: Transformers & Deep Learning – Context-aware models like BERT and GPT revolutionized NLP.



Comparison of NLP Approaches

Rule-Based NLP

- Uses predefined grammar rules.
- Simple and accurate for fixed tasks.
- Cannot learn or understand context.

Statistical NLP

- Uses probabilities from text data.
- More flexible than rule-based systems.
- Limited contextual understanding.

Machine Learning NLP

- Learns patterns from training datasets.
- Higher accuracy and adaptability.
- Requires large amounts of training data.

Modern NLP (Transformers)

- Understands context using deep learning.
- Human-like responses with high accuracy.
- Needs powerful computing resources.



Modern NLP Applications

Key Technologies:

- Deep Learning
- Transformers
- Attention Mechanism
- Large Language Models (GPT, BERT, Gemini)

Applications:

- ChatGPT, Google Translate, Siri, Alexa, Sentiment Analysis, Text Summarization, Spam Detection.



Future Trends & Conclusion

Future Trends:

- Multimodal AI
- Real-time Translation
- Personalized AI Assistants
- Healthcare & Education

Challenges:

- Bias, Privacy, Hallucinations and High Computing Cost.

Conclusion:

- NLP has evolved from simple handcrafted rules to intelligent language models capable of understanding and generating human language with remarkable accuracy.

Artificial Intelligence



THANK YOU

Ajay
Reddy